

AVINASH KUMAR SINHA

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Education

Manipal Institute of Technology (MIT Manipal)

Bachelor of Technology in Electronics and Communication

Sep. 2021 – May 2025

Manipal, Karnataka

Delhi Public School, Bokaro: XII(CBSE): 86.4%

2021

Delhi Public School, Bokaro: X(CBSE): 93.6%

2019

Professional Experience

SAIL, Bokaro

Jun 2024– Aug 2024

AI/ML Project Intern | Python, Thonny, YOLOv8, Google Collab

Bokaro, Jharkhand

- Developed a **Python**-based vehicle detection system using **YOLOv8**, enabling real-time license plate recognition.
- Integrated **speed estimation** from CCTV footage, enhancing traffic enforcement and safety through automated analysis.
- Delivered annual savings of **Rs.1.44 lakh+** by optimizing traffic monitoring and reducing manual intervention.

Projects

Banking Transactions Analysis | SQL, Excel

Jan 2025– May 2025

- Designed a normalized **MySQL database** to manage customer accounts, transactions, and banking metadata.
- Analyzed high-value and suspicious transactions using **JOIN, CASE, subqueries**, and **aggregation** techniques.
- Built data views and summary reports to identify inactive users, **transaction trends**, and potential **fraud indicators**.

Product Strategy for College Events App | Figma, Google Forms, Excel

Sept 2024– Dec 2024

- Conducted user interviews and analyzed **100+ survey** responses to identify student **pain points** in event discovery.
- Designed **MVP** and developed interactive **Figma wireframes** using **feedback** from early users and **peer testing** sessions.
- Tracked engagement via surveys, leading to a **35%** boost in student turnout and increased participation at events.

Hotstar Clone : End-to-End DevSecOps Deployment | HTML, CSS, Javascript

May 2024– Aug 2024

- Deployed a Dockerized **Hotstar-style frontend** on **AWS EKS**, securing images with **Docker Scout** and **OWASP** tools.
- Created **CI/CD pipelines** using **Jenkins** and **GitHub Actions**, with **SonarQube** for code quality and **static analysis**.
- Used **Trivy**, **DockerHub**, and **Kubernetes** tools to monitor and scan workloads in a production like setup.

Credit Card Fraud Detection System | Python, ML

Sept 2023– Dec 2023

- Built a fraud detection model using **Logistic Regression** and **Random Forest** on imbalanced credit card data.
- Preprocessed data with **Pandas**, applied **SMOTE** to balance classes, and performed basic feature engineering.
- Achieved over **90%** fraud detection accuracy using **Scikit-learn evaluation metrics** and **cross-validation**.

Technical Skills

Languages: SQL, C, C++, Python, Javascript, Java

Data Tools: Excel, Pandas, MySQL, Google Sheets,

Software Engineering Skills: Data Structures, Algorithms, Object-Oriented Programming (OOP), REST APIs

Analytics and Reporting: Data Cleaning, EDA, Transaction Pattern Analysis, Reporting Dashboards

Productivity and Design: Figma, Google Forms, Notion, Jira, Git

Soft Skills: Team Collaboration, Time Management, Problem Solving, Peer Feedback Integration

Certifications: GenAI Tools SkillNation, PowerBI OfficeMaster, , GenAI Powered Data Analytics Job Simulation (TATA), Data Analytics Professional (Google), Product Management (Doremon Den), Hotstar Clone Bootcamp(Google, Microsoft, AWS) , Fit in Deutsch 1 (Goethe Institut)

Positions of Responsibility

MIT, Manipal

Nov 2023

- Led a **15-member** team to plan and execute campus-wide events, ensuring seamless collaboration and timely delivery.
- Drove community engagement initiatives by establishing partnerships with local organizations, increasing event visibility.
- Managed end-to-end budgeting and resource allocation, to maximize cost-efficiency and ensure optimal resource utilization.

Achievements

- * Secured **State Rank 4** in National Talent Search Examination (**NTSE**) among 30,000+ candidates.
- * Achieved **AIR 217** in the Science and Mathematics Talent Examination (**SMTE**).